



مُعِين

صوت الأمان في أوقات الفطر

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INTRODUCTION

As accidents and emergencies rise, the demand for new tech innovations that can assist people in managing a life-critical situation. This app is intended to quickly render aid in case of fire, car crash, or emergency medical condition by automatically sending the geographical location to ambulance teams when the person in danger is at risk. It also offers a list of emergency numbers with the option to add few close ones to be notified automatically in case of an accident. The application is characterized by its ability to work without an internet connection which guarantees effectiveness even in sites where there is weak or no network. It includes the feature of rapid entry in cases of assault whereby the user can easily activate the application by pressing a specific button in the phone, and the application starts recording audio and sending geographical location coordinates to the police. Furthermore, the application will indicate the nearest police station or hospital to the accident site for ensuring quick response time. It will review the importance of this application its role in personal safety, and the challenges it might face in development.

ABSTRACT

This document outlines a mobile application aimed at enhancing emergency responses in urgent situations such as accidents or medical emergencies. The application offers instant assistance via automated voice guidance and hands-free interaction, enabling rapid access to support.

Main features comprise voice command recognition to identify emergencies and notify authorities with accurate location information, as well as audio instructions for users until help arrives.

The application, accessible on both iOS and Android, strives to boost usability and accessibility, thereby enhancing emergency response times and individual safety.

CHAPTER ONE

1.1 PROBLEM DEFINITION

Many people struggle to get help in emergencies, which can be dangerous, Fear and injury may stop them from calling for help. This app offers a simple and easy solution by pressing a specific button on the phone to send an emergency alert quickly, reducing delays and ensuring faster help.

1.2 OBJECTIVES

- 1- Provide real-time voice guidance to assist users during emergencies with step-by-step audio instructions.
- 2- Enable hands-free interaction through voice commands and a shortcut button for quick activation.
- 3- Automate emergency alerts by detecting incidents and sending location-based notifications to emergency contacts.
- 4- Ensure offline functionality to allow operation without an internet connection.
- 5- Enhance safety and accessibility by offering discreet activation, location-based assistance, and a user-friendly interface.

1.3 EXPECTED OUTCOME

- 1- Faster and more efficient EMERGENCY response.
- 2- Automatic location tracker for a faster assistance.
- 3- reliable performance in various emergency situations.

1.4 SCOPE OF THE OBJECT

The Ma'een project ensures secure login, protects personal data, and manages medical history for emergencies. It enables instant communication with emergency services and personal contacts, supports GPS tracking for location sharing, and includes language recognition for faster response.

The app features voice command support on various devices, educational first aid tutorials, an emergency siren activated via the phone's home button, and offline functionality for use in low-network areas. Its user-friendly interface ensures accessibility for all.

Out of Scope:

- 1- No medical diagnoses or consultations.
- 2- No direct integration with hospital systems.
- 3- No video calls with emergency responders.

1.5 Fasibility Analysis

1- Requirements

Hardware: Smartphone with GPS, microphone, speaker, and optional internet.

Software: Mobile app (Android/iOS), GPS integration, offline mode, emergency contact database.

2- Functions

Emergency Assistance: Sends user location to emergency services.

Audio Guidance: Provides voice instructions for different emergencies.

Contact Alerts: Notifies predefined emergency contacts.

Offline Support: Works without internet for guidance and alerts.

Quick Activation: Trigger emergency mode via a button press.

Location Services: Identifies nearby hospitals or police stations.

3-Restrictions

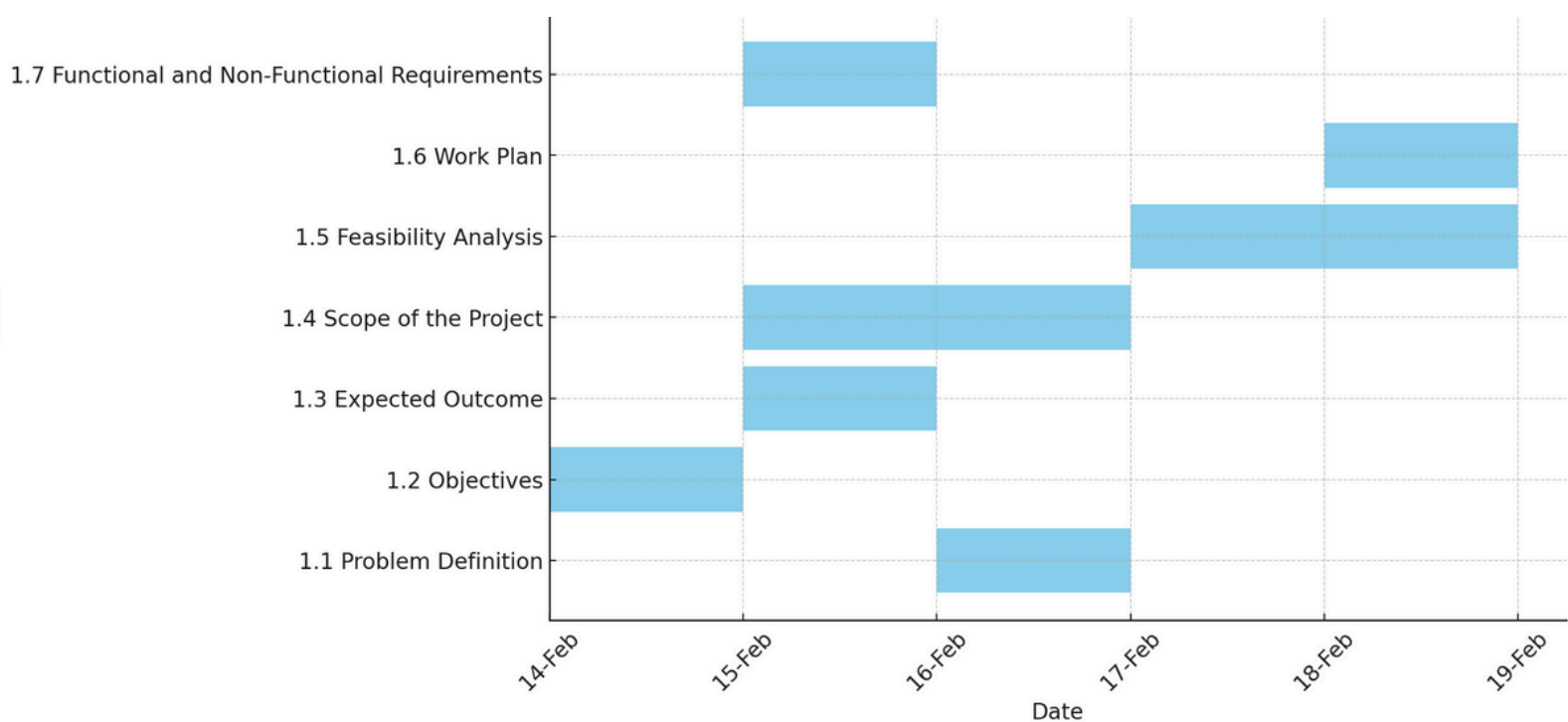
Technical Limitations: GPS accuracy, offline messaging constraints.

Privacy Concerns: Secure storage and transmission of user data.

Legal Compliance: Adherence to local emergency service protocols.

1.6 work plan

TASK	START	END
1.1 Problem definition	16-Feb	17-Feb
1.2 Objectives	14-Feb	15-Feb
1.3 Expected outcomes	15-Feb	16-Feb
1.4 Scope of the project	15-Feb	16-Feb
1.5 Feasibility Analysis	15-Feb	17-Feb
1.6 Work plan	17-Feb	19-Feb
1.7 Functional & Non-Functional requirements	15-Feb	16-Feb



1.7 FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Functional requirements:

1-Voice command recognition

It can recognize voice commands such as when there is an accident, a fire, an attack, etc.

2-Automatic emergency call

The application can directly contact the police, ambulance, or civil defense and send the exact geographic location quickly

3-Send alert to specified contacts

It sends an emergency notification with the geographic location sent to the people specified in advance

4-Audio guides

It provides voice instructions according to the type of emergency

Non-functional requirements:

1-Performance

It works on iOS and Android without the need to connect to the Internet and supports more than one language.

2-Response

It responds to voice commands in seconds

3-Ease of use

The application works easily and clearly to suit all age groups

CHAPTER TWO

2.1 BACKGROUND

These days, with the increasing number of emergencies such as car accidents, fires, and health crises, quick action and sound decision-making are essential to save lives and minimize damage. However, many people struggle to cope with these situations due to stress or an inability to quickly access emergency services. Therefore, our app was developed to be a smart voice assistant specifically designed for such situations. It guides the user verbally, helping them take the correct action based on the type of emergency they are facing. It also automatically sends the location to emergency services without the need for an internet connection, ensuring a faster response and contributing to saving lives.

2.2 COLLECTION OF FACTS

This is an electronic questionnaire designed to gather opinions on emergency applications and their essential features. The survey targeted individuals aged 17 and above. The questionnaire included the following questions:

1. Gender?
2. Age?
3. Have you ever faced an emergency situation where you wished for an app that could help you react quickly?
4. Would you prefer the app to automatically send your location in case of an accident?
5. Would you prefer receiving voice instructions for emergency guidance instead of reading text instructions?
6. Would you like the app to include a direct call button for emergency services such as the police or ambulance?
7. What feature do you consider the most important in an emergency app?

• We collected responses from the questionnaire, and the results were as follows:

1. Gender Distribution:

- 31% of respondents were male.
- 69% of respondents were female.

2. Age Distribution:

- 12-25 years old: 49%
- 26-35 years old: 26%
- 36-45 years old: 19%
- 46-55 years old: 6%

3. Need for an Emergency App:

- 74% of respondents have experienced an emergency where they wished for an app to help them react quickly.
- 26% have not faced such a situation.

4. Automatic Location Sharing:

- 100% of respondents preferred the app to automatically send their location in case of an accident.
- 0% did not find it necessary.

5. Voice Instructions vs. Text Instructions:

- 95% of respondents preferred voice instructions for emergency guidance instead of reading text.
- 5% preferred reading instructions.

6. Direct Emergency Call Feature:

- 98% of respondents supported having a direct call button for emergency services.
- 2% did not find it necessary.

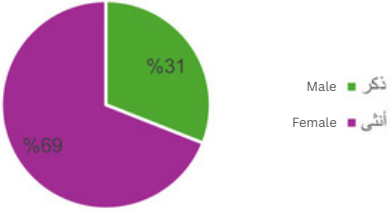
7. Most Important Feature in an Emergency App:

- 34% prioritized automatic location sharing with emergency services.
- 31% considered providing voice instructions to handle the situation the most crucial feature.
- 35% found offline functionality the most important.

2.3 THE QUESTIONNIRE RESULTS

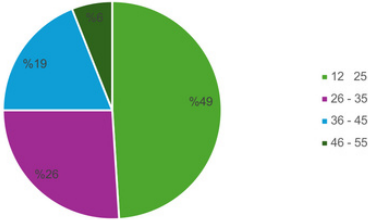
Gender:

الجنس:



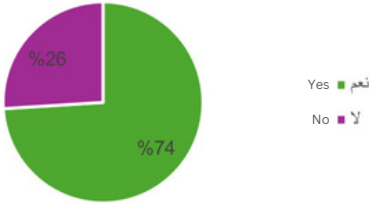
Age:

العمر:



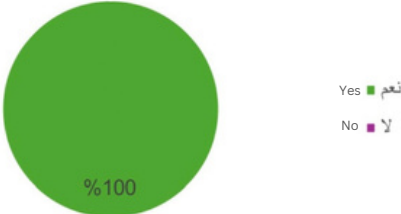
هل سبق لك أن واجهت موقفًا طارئًا تمنيت فيه وجود تطبيق يساعدك على التصرف بسرعة؟

Have you ever faced an emergency situation in which you wished there was an app that would help you act quickly?



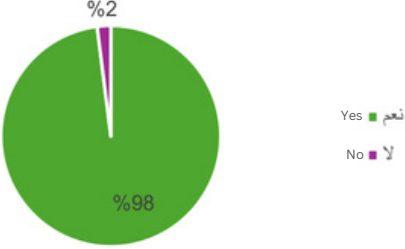
هل تفضل أن يكون التطبيق قادرًا على إرسال موقعك تلقائيًا عند وقوع حادث؟

Would you prefer the app to be able to automatically send your location when an accident occurs?



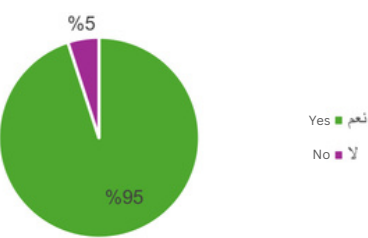
هل تفضل وجود زر اتصال مباشر بالشرطة أو الإسعاف داخل التطبيق؟

Do you prefer a direct call to the police or ambulance button within the application?



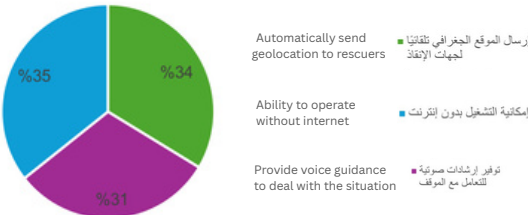
هل تفضل الحصول على إرشادات صوتية لمساعدتك أثناء الطوارئ بدلاً من قراءة التعليمات؟

Would you rather have voice instructions to help you during an emergency instead of reading the instructions?



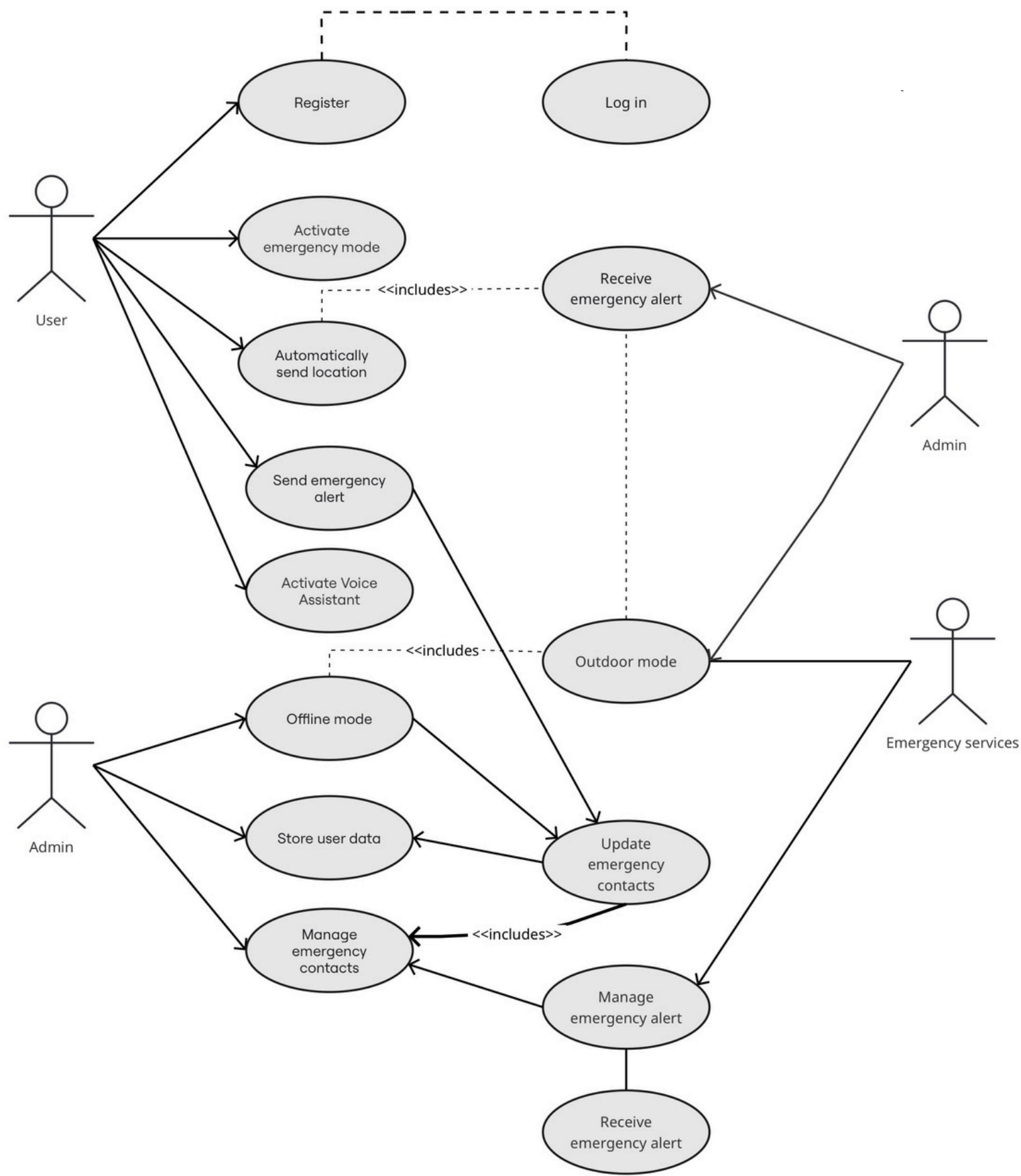
ما الميزة التي تراها الأكثر أهمية في تطبيق طوارئ؟

Which feature do you think is most important in an emergency app?



CHAPTER THREE

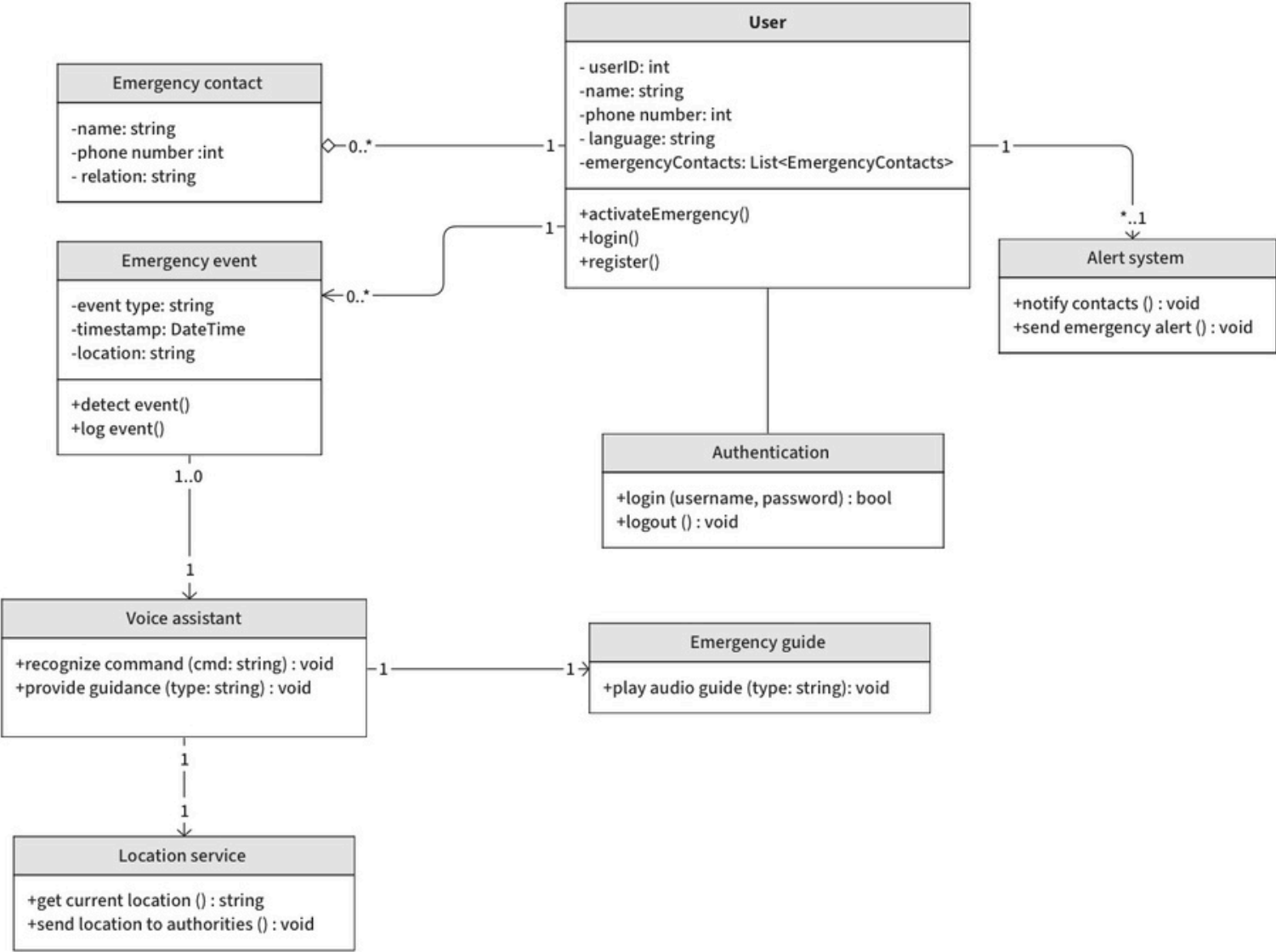
3. 1 USE CASE



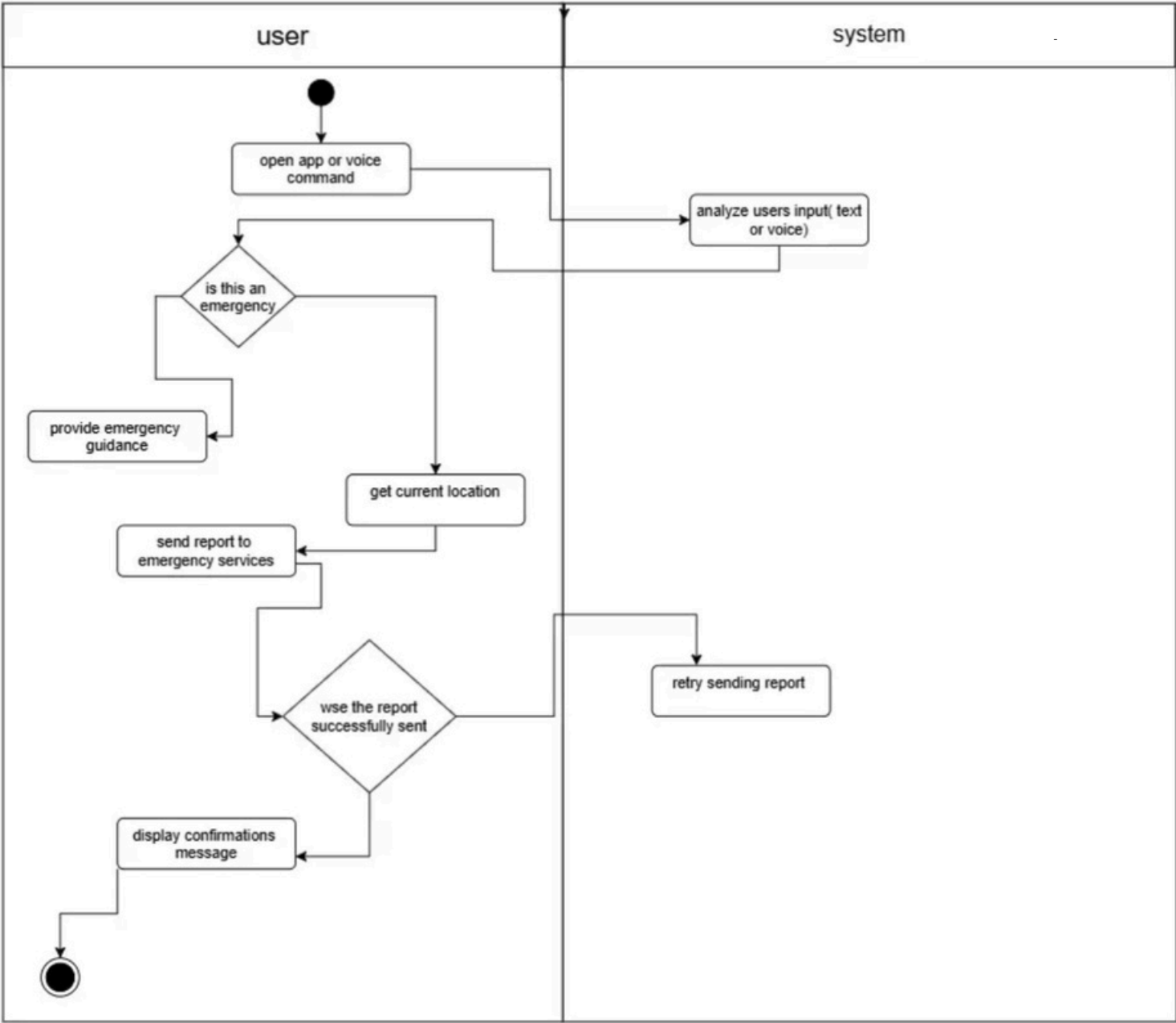
3.2 USE - CASE DESCRIPTIONS

Use Case name: مُعِين	ID: 1	Importance Level: High
	Use case type: Essential	
Primary actor: caller		
Caller: Emergency Services: receive location and information quickly. Admin: Ensures the system functions properly and updates emergency info. Admin: Admin activates the emergency mode in a critical situation to initiate app assistance.		
Brief Description: This use case describes how the user activates the emergency voice assistant to receive real-time audio guidance and send location to emergency services, even without internet such as SMS fallback.		
Trigger: The user faces an emergency (e.g., accident, fire, medical case) and needs immediate help		
Type: Essential Flow		
Relationships: Include: The user sends location automatically. Include: The user calls emergency services. Extend: The user interacts via voice commands. Extend: Send additional health info to services if available.		
Normal flow of event: 1. Activate emergency mode by voice or button. 2. A Voice assistant starts guiding the user. 3. System automatically sends location to the nearest emergency service. 4. Emergency service receives the location and responds.		
Sub-flows: 1. The user issues a voice command to trigger emergency mode. 2. System works offline if no internet is available. 3. Admin updates and manages guidance instructions.		

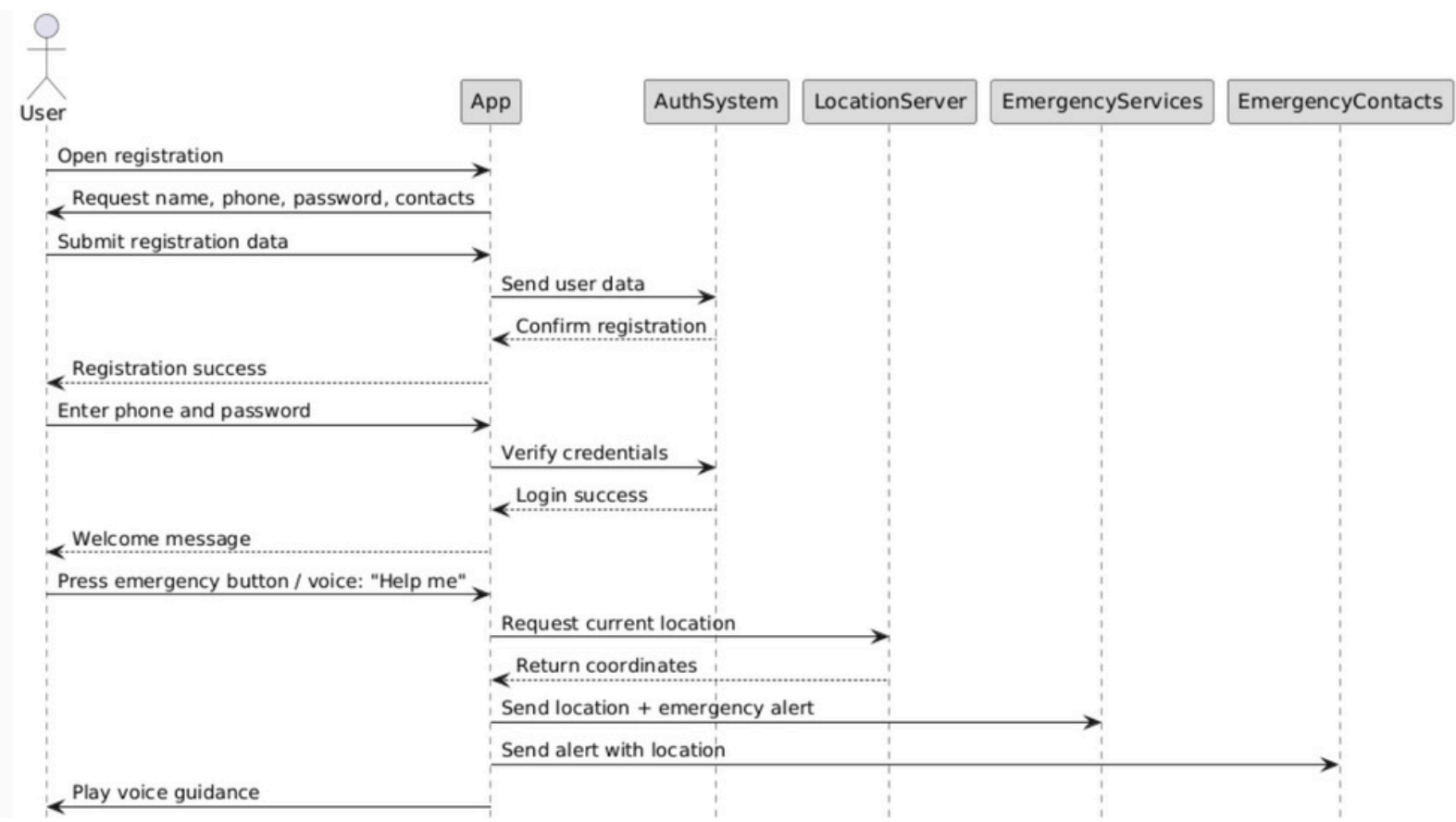
3.3 CLASS DIAGRAM



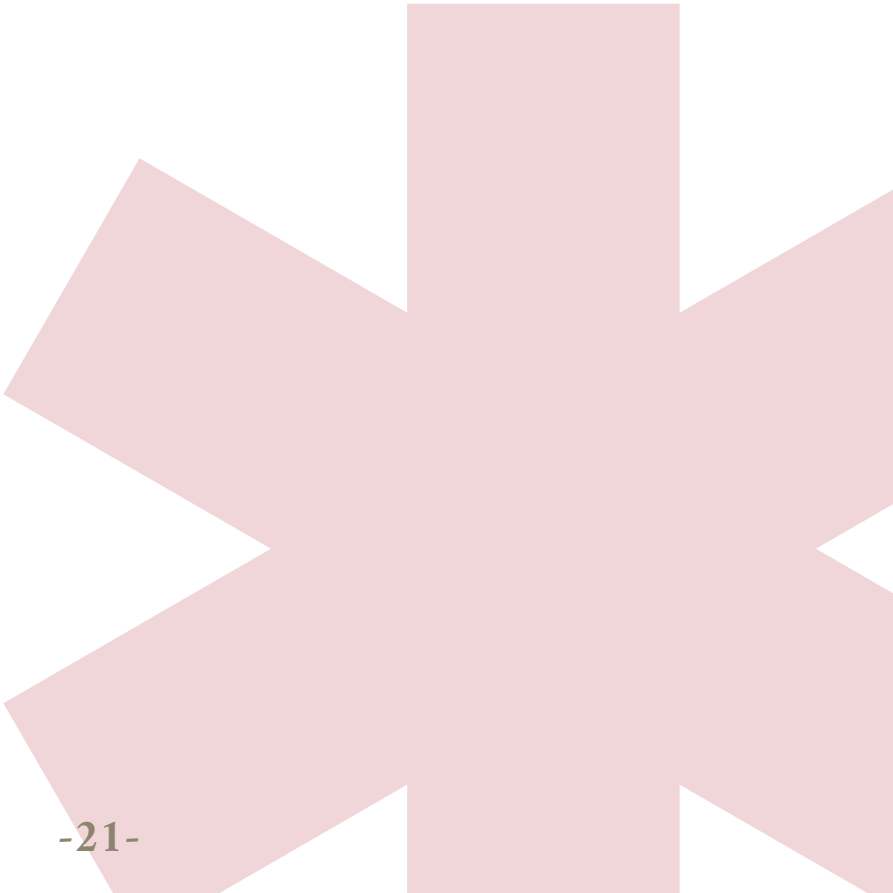
3.4 ACTIVITY DIAGRAM



3.5 SEQUENCE DIAGRAM



CHAPTER FOUR



4.1 DESIGN STRATEGIES:

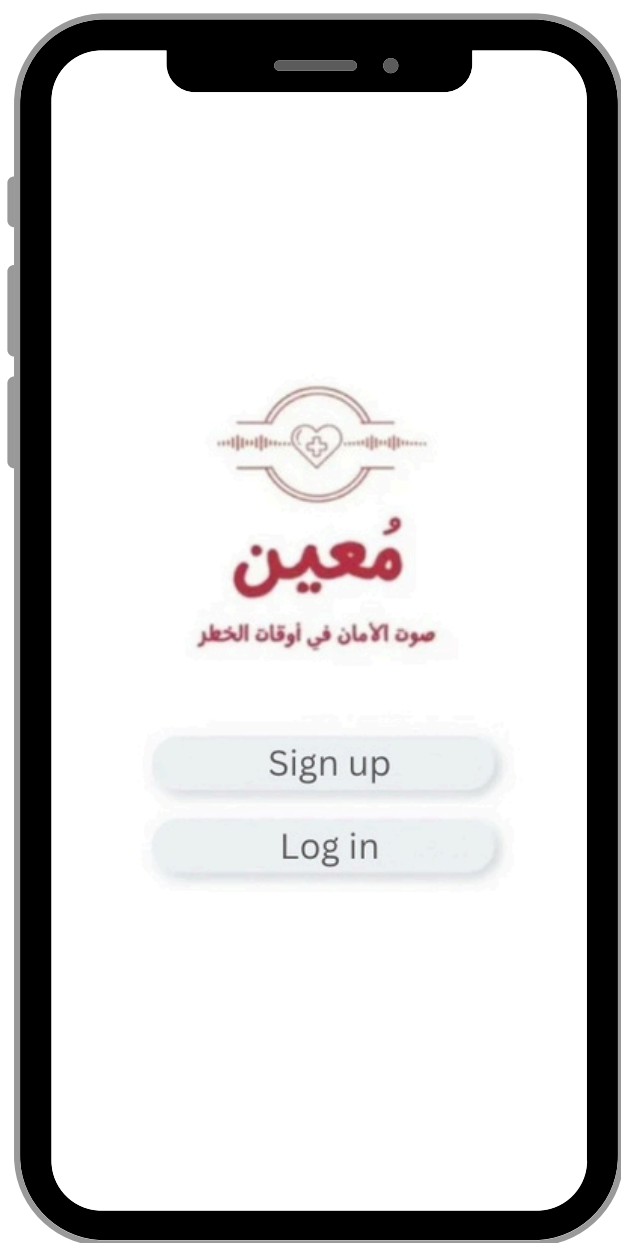
User-Centered Design: Ensure simplicity and accessibility for all age groups.

Fast Interaction: Minimize steps to reach emergency help.

Offline Functionality: App works even without internet.

Security & Privacy: Protect user data and location info.

4.2 USER INTERFAC



4.2 USER INTERFAC

Sign up

Name

Phone number

National id

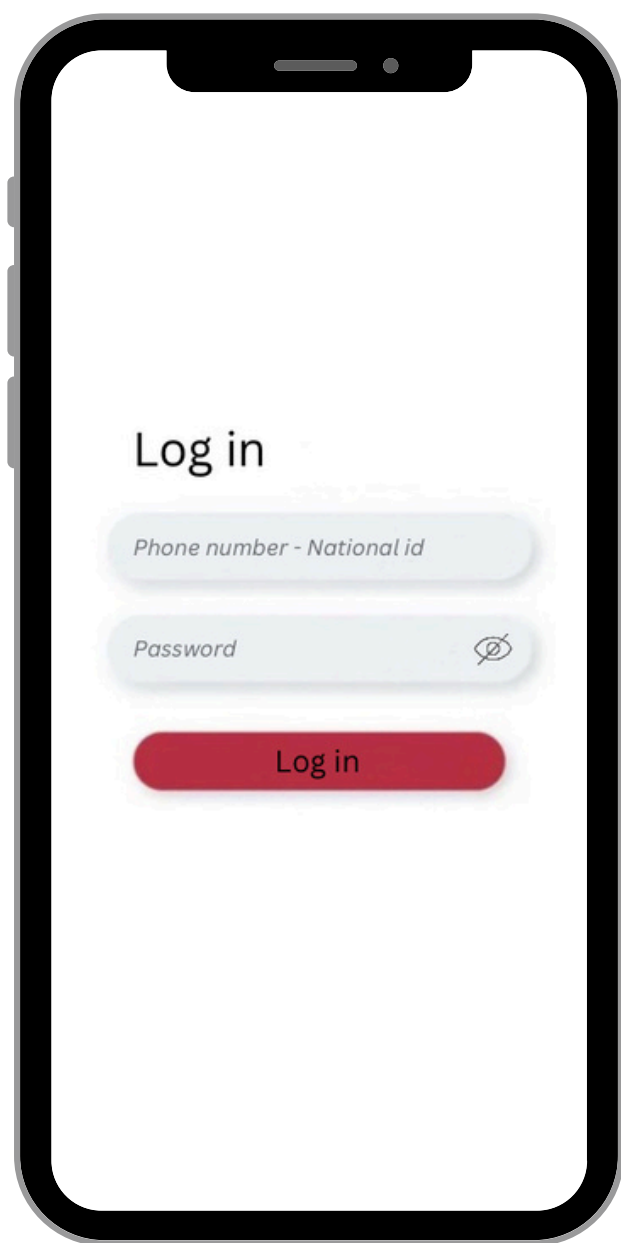
Birth date

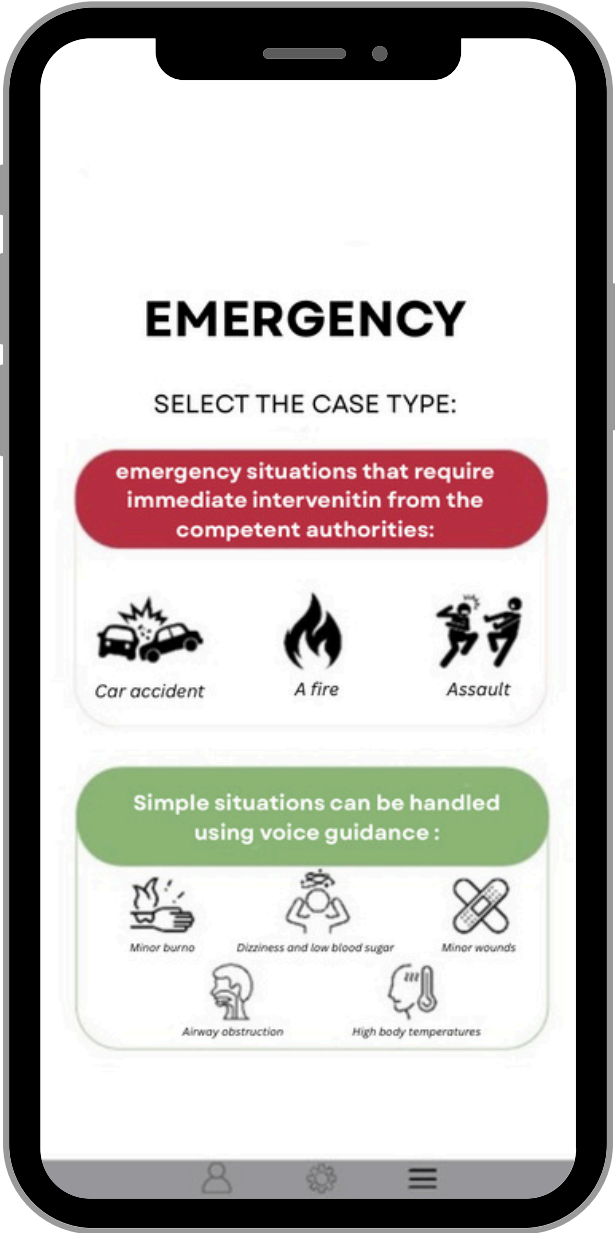
Password

Confirm password

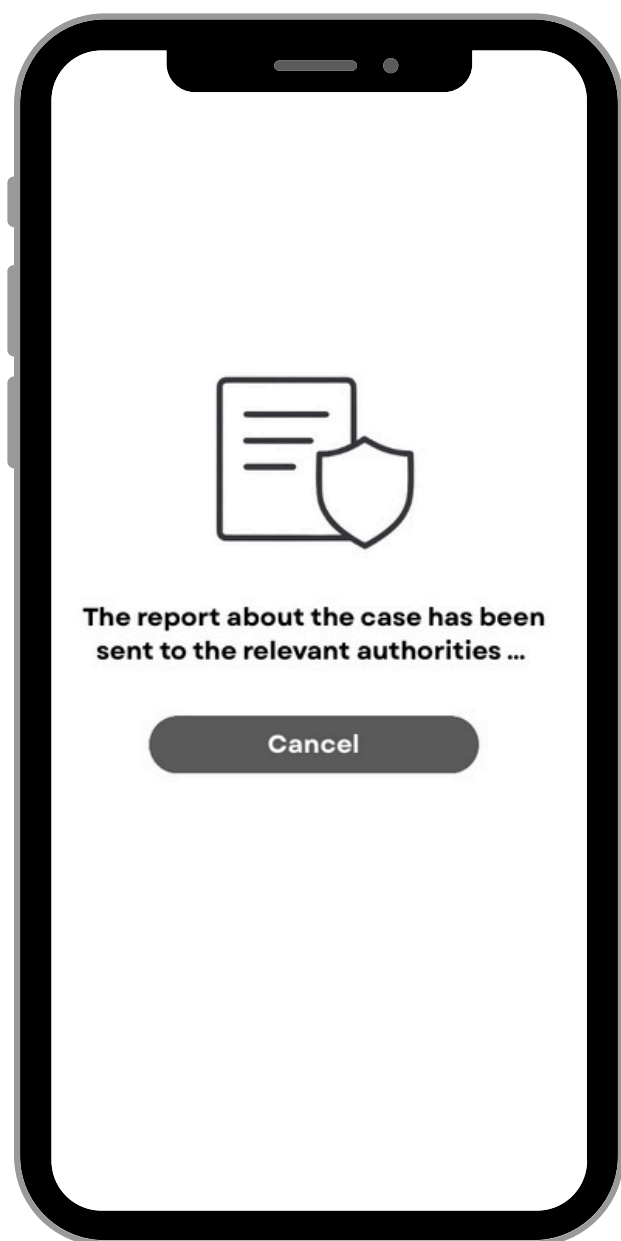
Sign up

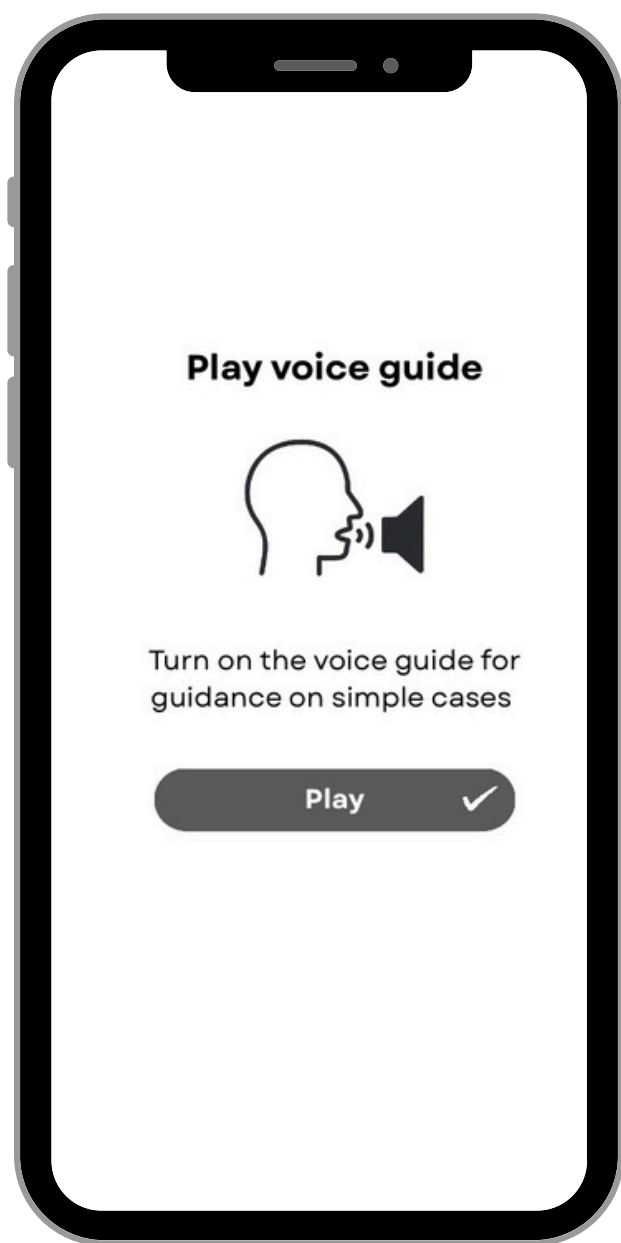
4.2 USER INTERFAC



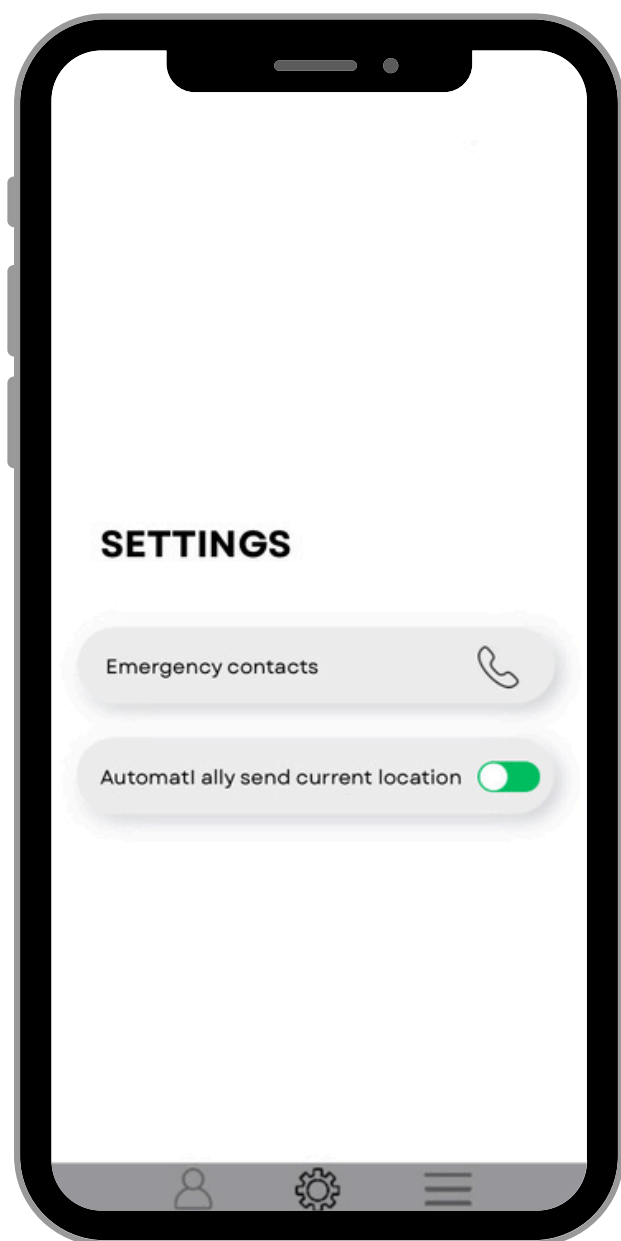


4.2 USER INTERFAC

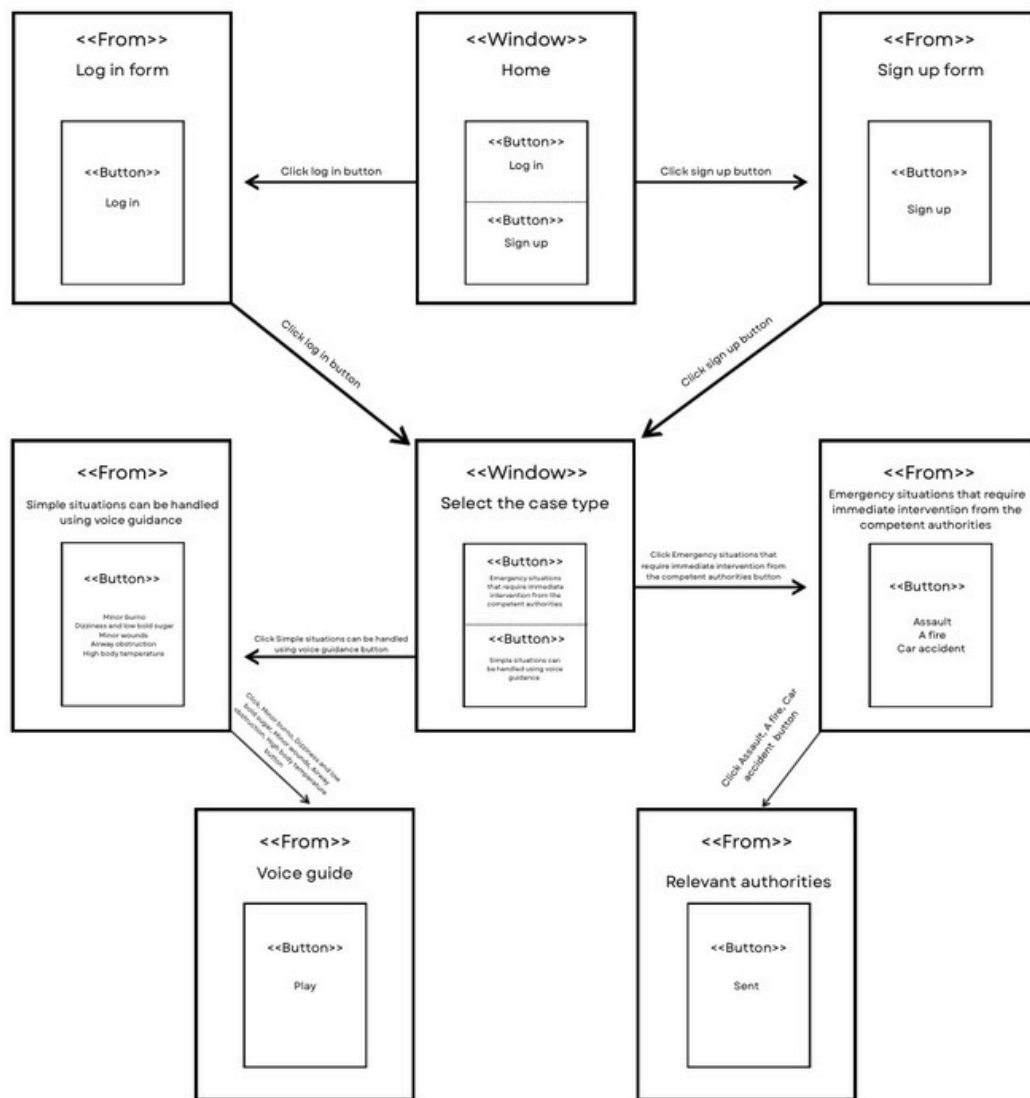




4.2 USER INTERFAC



4.3 WINDOWS NAVIGATION DIAGRAM



4.4 interface design prototyping:

In interface design prototyping we used a windows navigation diagram

CHAPTER FIVE

5.1 CONCLUSION

This project was created to meet the growing need for smart solutions that match today's fast technological progress.

It aims to improve how we respond to emergencies in a safe and quick way. We hope that "Mu'een" will be a useful tool that helps different emergency services work better together and leads to a more advanced and connected emergency system in the future.

5.2 Future Work

Adding a feature that allows users to send media (videos or photos) explaining the emergency situation, so the system can guide the user on the best way to handle it.

Direct integration with official emergency services, enabling immediate communication between emergency teams and the user, ensuring a faster and more effective response.

Designing interfaces and options that make the system easier for people with disabilities to use in emergency situations, including features specifically designed for disabilities (visual, hearing, mobility, etc.)